

ABDULRAHMAN BINFARIS, PhD

Academic Curriculum Vitae | Mechanical Engineering, Smart Materials & Systems

United States | 02binfaris@cua.edu

Google Scholar: scholar.google.com/citations?user=2AhsGuIAAAAj

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RESEARCH PROFILE

Mechanical engineering researcher with experience in smart materials, carbon nanotube (CNT) sensing, structural health monitoring, experimental validation and integrated engineering systems. Research includes NASA-funded multi-sensing platforms and smart hull prototypes. Academic experience includes research leadership, adjunct instruction, graduate teaching and mentoring doctoral researchers.

RESEARCH INTERESTS

Smart and multifunctional materials; CNT fibers and composites; structural health monitoring; embedded sensing; experimental mechanics and damage detection; adaptive structures; renewable energy and power systems; systems integration.

EDUCATION

PhD, Mechanical Engineering - Smart Materials & Systems

The Catholic University of America, Washington, DC

MS, Mechanical Engineering - Renewable Energy and Cleaner Production

The Catholic University of America, Washington, DC

BS, Mechanical Engineering - Aerospace

The Catholic University of America, Washington, DC

ACADEMIC AND RESEARCH APPOINTMENTS

Principal Research Scientist | The Catholic University of America

December 2024 - Present

Postdoctoral Fellow | The Catholic University of America

August 2023 - December 2024

Research and Development Director | Intelligent Materials Laboratory, CUA

June 2020 - December 2024

Led smart materials research, integrated multi-sensing development and prototype fabrication and testing. Developed Python-based software for damage detection and sensor-data analysis in NASA-funded research.

Graduate Research Assistant | The Catholic University of America

August 2016 - May 2020

TEACHING AND MENTORING

Adjunct Professor | Trinity Washington University, Washington, DC

January 2022 - December 2022

Delivered instruction in power and environmental topics, incorporating renewable energy applications and interactive learning.

Graduate Teaching Assistant | The Catholic University of America

September 2017 - May 2020

Research mentoring: Mentored nine doctoral researchers in research and engineering activities.

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SELECTED RESEARCH AND APPLIED DEVELOPMENT

NASA-funded smart materials and integrated sensing research | CUA

Developed integrated multi-sensing platforms and Python-based methods for sensor analysis and damage detection. Work connects smart material behavior, measurement and structural assessment.

Smart hulls for underwater unmanned vehicles | CUA

Led research on UUV smart hulls, including prototype fabrication and testing. Related research experience includes CNT-based sensing, smart composites and structural monitoring.

Portable optical water measurement | AccuStrata

Supported systems engineering and validation for a \$1.4 million Phase II development program targeting TRL 7. This amount describes the program, rather than an individual grant award.

Hybrid power systems | ATSC

Directed engineering for an approximately 1 MW hybrid power system within a Department of Defense program, covering load analysis, generation, energy storage, distribution and supervisory control.

RESEARCH METHODS AND TECHNICAL CAPABILITIES

Experimental work: strain, fatigue, vibration and temperature testing; resistance and impedance measurements; calibration; prototype development; verification and validation.

Tools and analysis: Python, LabVIEW, National Instruments data acquisition, finite element analysis and vector network analysis.

Systems: requirements development, sensor integration, optical and electrical measurement, power systems and multidisciplinary engineering coordination.

APPLIED ENGINEERING AND RESEARCH LEADERSHIP

Chief Operating Officer (COO) and Co-Founder | IntSensing

October 2023 - Present

Lead engineering and R&D coordination for integrated sensing and structural health monitoring projects, connecting research development with systems implementation.

Senior Principal Systems Engineer (Consultant) | AccuStrata

May 2025 - May 2026

Supported optical measurement development, systems integration and experimental validation.

Power Systems Director | ATSC

August 2020 - September 2023

Directed multidisciplinary power systems engineering and integration for a defense program.

ACADEMIC SERVICE

Member, School of Engineering Academic Advisory Committee, The Catholic University of America | January 2018 - January 2020

PUBLICATIONS AND SUPPORTING ACADEMIC RECORD

Publication profile: [Google Scholar - Abdulrahman Binfaris](#).

This concise application version is pending a verified full publications and grant-award list. Individual publication citations, funding awards and detailed teaching portfolios are not included.

LANGUAGES

Arabic: native/bilingual proficiency | English: full professional proficiency